



CHARLES UNIVERSITY
FACULTY OF SOCIAL SCIENCES
Institute of Economic Studies

Tools for Modern Macroeconometrics

Term Paper: Macroeconomic Forecasting and Shock Propagation in Germany

Václav Šmíro

May 28, 2026

Contents

1	Introduction	2
1.1	Motivation and Objectives	2
1.2	Historical and Economic Context	2
2	Univariate Model	3
2.1	Data Sources and Transformations	3
2.2	Model Selection and Residual Diagnostics	4
2.3	Univariate Forecasts	6
3	Multivariate Models: Vector Autoregressive (VAR) Analysis	8
3.1	Exogenous Variables and Monetary Policy Proxy	8
3.2	VAR Specification and Diagnostics	9
3.3	Identification and Impulse Responses	10
3.4	VAR Forecasts	13
4	Bayesian VAR and Sign Restrictions	14
4.1	Bayesian Estimation	14
4.2	BVAR Sign Restrictions and Structural Identification	15
4.3	BVAR Forecasts	16
5	Models with Trend Variables	17
5.1	VAR in Levels Specification and Analysis	17
5.2	Vector Error Correction Model (VECM)	19
6	Conclusion	20
	Appendix	21
	References	23

1 Introduction

1.1 Motivation and Objectives

The main goal of this term paper is to create forecasts of real Gross Domestic Product (GDP) and inflation for Germany and to identify how macroeconomic shocks spread through the economy. This paper constructs short- to medium-term predictions starting in 2026Q1 and extending into subsequent years. To better understand the data-generating processes, the empirical analysis follows: First, univariate time series techniques using the Autoregressive Integrated Moving Average (ARIMA) framework are applied to historical indicators. Second, this paper develops multivariate frameworks to capture external shocks. We estimate a series of Vector Autoregressive (VAR) models that incorporate two exogenous transmission channels: monetary policy shocks, captured by a synthesized short-term interest rate that accounts for the Zero Lower Bound (ZLB) period via a shadow rate, and global commodity supply shocks, represented by the spot price of Brent crude oil. The multivariate analysis including recursive Structural VARs (SVAR) identified through Cholesky ordering, structural models with explicit short-run parameter restrictions on central bank behavior, Bayesian VAR (BVAR) models and BVAR models identified via structural sign restrictions.

1.2 Historical and Economic Context

For the proper specification and interpretation of the multivariate time series models, it is necessary to understand the historical development of the German economy. From the macroeconomic perspective, Germany is the largest and most industrialized export-oriented economy in the European Union. At the same time, the country is a major net importer of primary commodities. For this reason, global oil prices were selected as a key variable in this model. Given Germany's specific structural characteristics as a commodity-dependent exporter, the empirical results of energy price shocks on German macroeconomic variables can provide highly interesting insights and illustrate how external supply shocks are transmitted to a major industrial economy ([Deutsche Bundesbank, 2023](#)).

From Reunification to the Eurozone (1990–2007)

The macroeconomic development of Germany in the modern era was fundamentally affected by the reunification in the year 1990. This unprecedented process represented not only a structural economic shock but also social challenge. Interestingly, opinion polls in the late 1980s showed that East and West Germans perceived each other as foreign and did not regard themselves as a single nation anymore ([Jarausch, 1994](#)). Despite this social divergence, the rapid integration of the East German economy demanded massive fiscal transfers and caused an initial boom in consumption, which generated significant inflationary pressures.

In response to this development, the independent Deutsche Bundesbank applied a strictly tight monetary policy to anchor inflation expectations ([Sinn, 2002](#)). At the time of the Euro adoption in 1999, when the monetary policy was delegated to the European Central Bank (ECB), Germany suffered from structural rigidities and low growth of real GDP, which led to the label “the sick man of Europe” ([Dustmann et al., 2014](#)). However, the following implementation of the Hartz labor market reforms (2003–2005) increased the competitiveness of the economy. This resulted in strong export-led GDP growth and stable HICP inflation in the period before the global financial crisis.

The Great Recession and the Zero Lower Bound (2008–2019)

The realization of the Great Financial Crisis in 2008 represented a massive negative demand shock. The export-heavy German economy experienced a sharp contraction in real GDP. At the same time, global oil prices dropped significantly, which pushed the headline inflation down. The ECB reacted by aggressive cuts in the key interest rates to support aggregate demand ([European Central Bank, 2010](#)).

During the later European Sovereign Debt Crisis, the standard transmission mechanism was impaired as the ECB reached the Zero Lower Bound (ZLB). To stimulate the Eurozone economy, the central bank introduced unconventional monetary policy measures, including Quantitative Easing (QE). In this environment, standard short-term interest rates were constrained by the zero floor and failed to capture the true expansionary stance of monetary policy. To address this empirical challenge, a shadow interest rate is incorporated into our multivariate models. As noted by [Krippner \(2015\)](#) and [Wu and Xia \(2016\)](#), the shadow rate serves as an unconstrained proxy variable for the monetary policy stance when conventional policy rates are stuck at or below zero. Despite this massive monetary expansion, the German HICP inflation remained systematically below the 2% target of the ECB throughout the 2010s ([Deutsche Bundesbank, 2019](#)).

Pandemic Shock and the Return of Great Inflation (2020–Present)

The start of the COVID-19 pandemic in 2020 caused a temporary but massive drop in real GDP due to economic lockdowns and disrupted supply chains. However, the most severe structural shock occurred in early 2022 following the Russian aggression in Ukraine. Because of the high dependency of the German industrial sector on imported energy, the economy suffered a severe shocks ([Deutsche Bundesbank, 2022](#)). The exponential surge in global oil and gas prices acted as a exogenous supply shock, which rapidly translated into the domestic price level.

This dynamic development triggered the “Great Inflation” of the 2020s, and the German HICP inflation reached double digit values. In reaction to the inflation shock, the ECB sharply increased interest rates and exited the ZLB regime. The combination of the negative energy price shock and the restrictive monetary policy has slowed down the growth of German real GDP.

2 Univariate Model

2.1 Data Sources and Transformations

To establish the baseline univariate models we use for economic activity, we use *Real Gross Domestic Product for Germany*, obtained from the Federal Reserve Bank of St. Louis Economic Data (FRED) database. The series is denominated in millions of euros, sampled at a quarterly frequency, and is seasonally adjusted. The historical sample period for GDP spans from the first quarter of 1991 through the final quarter of 2025.

For the price level, the main variable is the *Harmonized Index of Consumer Prices (HICP)*, extracted from the European Central Bank (ECB) Data Portal. To ensure empirical robustness, the national *Consumer Price Index (CPI)* from FRED is additionally added as an alternative price metric. Both price indicators are sampled at a monthly frequency and are initially obtained in their seasonally unadjusted forms. The evaluation period for the HICP data extends from January 1996 to February 2026, while the longer national CPI series spans from January 1975

to March 2025.

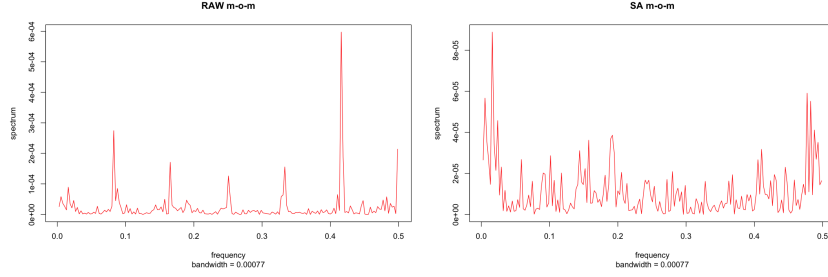


Figure 1: Periodogram and Spectral Density Analysis

To confirm the presence of seasonal variations in the unadjusted inflation data, a spectral analysis was conducted. Figure 1 displays the periodogram and the estimated spectral density, contrasting the raw unadjusted series with an adjusted series. The spikes observed at the seasonal frequencies provide evidence that seasonality is a dominant feature of the unadjusted inflation series. Consequently, seasonal adjustment is performed using **TRAMO-SEATS** via the **RJDemetra** package to filter out the seasonal noise.

Before model estimation, the data must be transformed to achieve stationarity. For all variables, logarithmic transformations are implemented to stabilize historical variance and deal with exponential growth effects. Because the log-levels of these macroeconomic series exhibit trends, differencing is required. For the quarterly GDP series, two specifications are examined: the first difference of logarithms, capturing quarter-on-quarter (q-o-q) growth rates, and the fourth difference of logarithms, capturing year-on-year (y-o-y) growth rates. For the monthly inflation indicators, we compute the first difference of logs to obtain month-on-month (m-o-m) inflation rates, and twelve-month differences to represent year-on-year (y-o-y) inflation rates.

To verify that these transformations successfully induced stationarity, Augmented Dickey-Fuller (ADF) unit root tests were applied across all series. The null hypothesis of a unit root is rejected at the 1% confidence level for both the q-o-q and y-o-y log-differenced GDP series. For the price variables, the ADF test rejects the unit root null hypothesis at the 5% confidence level. A structured summary of the specific variables, frequencies, sample spans, and applied transformations is detailed in Table 1.

Table 1: Data Specifications for Univariate Models

Variable	Frequency	Sample Period	Transformation
Real GDP	Quarterly	1991Q1 – 2025Q4	$\Delta \ln(GDP)$ (q-o-q), $\Delta_4 \ln(GDP)$ (y-o-y)
HICP	Monthly	1996M01 – 2026M02	$\Delta \ln(HICP)$ (m-o-m), $\Delta_{12} \ln(HICP)$ (y-o-y)
CPI	Monthly	1975M01 – 2025M03	$\Delta \ln(CPI)$ (m-o-m), $\Delta_{12} \ln(CPI)$ (y-o-y)

2.2 Model Selection and Residual Diagnostics

The optimal lag structure for the ARIMA processes is determined using Box-Jenkins step-by-step approach, combining visual inspection of Autocorrelation Functions (ACF) and Partial Autocorrelation Functions (PACF) with information criteria minimization.

For the year-on-year GDP series, visual inspection of the sample ACF and PACF plots in Figure 2 reveals an extensive number of statistically significant spikes that exceed the conventional

95% confidence bounds.

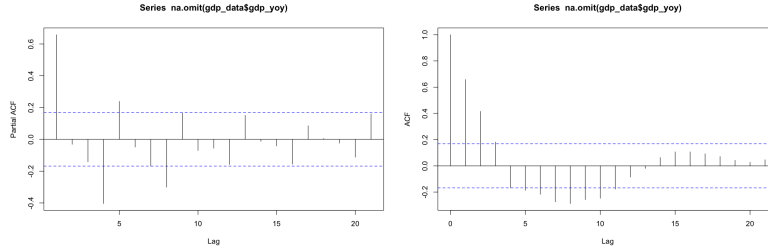


Figure 2: ACF and PACF of Year-on-Year (y-o-y) Real GDP

The sample partial autocorrelations reveal highly prominent, significant spikes at multiple lags, most notably at lags 1, 4, 5, and 8. On the basis of systematic optimization via the Akaike Information Criterion (AIC), an $ARMA(1, 4)$ specification is selected as the optimal model for the differenced y-o-y GDP series. To evaluate the statistical adequacy of this selection, a Ljung-Box portmanteau test is applied to the estimated residuals. The test yields a p-value of 0.3012, thereby failing to reject the null hypothesis of no serial correlation. This confirms that the $ARMA(1, 4)$ have residuals that behave as a white noise process.

In contrast, the quarter-on-quarter (q-o-q) GDP growth series exhibits different serial correlation properties. As illustrated in Figure 3, the ACF and PACF plots show no statistically significant lags, pointing toward a white noise process.

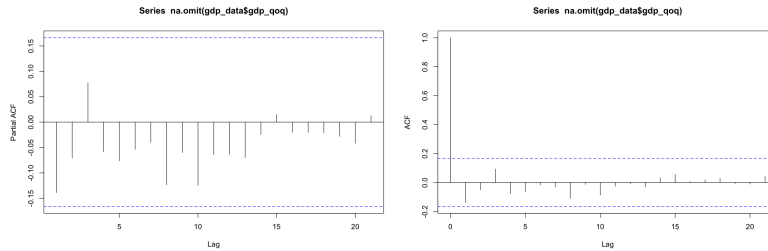


Figure 3: ACF and PACF of Quarter-on-Quarter (q-o-q) Real GDP

Nevertheless, model selection based on the AIC criterion favors a $MA(1)$ specification. The estimated $MA(1)$ coefficient is found to be -0.1515 , which is marginally significant with an empirical t-statistic of approximately -1.77 . The inclusion of a statistically significant constant term (0.0028) reflects the long-term positive average growth rate of the German economy over the post-reunification era. Diagnostic testing via the Ljung-Box test confirms model fit, returning a p-value of 0.5530.

For the year-on-year inflation series, both the HICP and national CPI measures display a very slow decay in their empirical ACFs. This high degree of persistence is expected, given that year-on-year values are constructed from overlapping 12-month periods, inducing a high degree of artificial moving-average smoothing. The PACF plots shows significant spikes at lag 13 and lag 25, suggesting that a degree of unmitigated seasonal persistence or long-memory behavior. This high persistence frequently emulates a non-stationary random walk process. Attempts to resolve this by executing a second differentiation or estimating a formal Seasonal ARIMA (SARIMA) model failed to have better empirical fits. Furthermore, incorporating a specialized dummy variable to control for the extreme volatility observed during the COVID-19 pandemic

did not the persistence parameters. Ultimately, minimizing the global AIC criterion leads to the selection of an $ARIMA(1, 1, 4)$ process as the most adequate representation for the y-o-y price indicators.

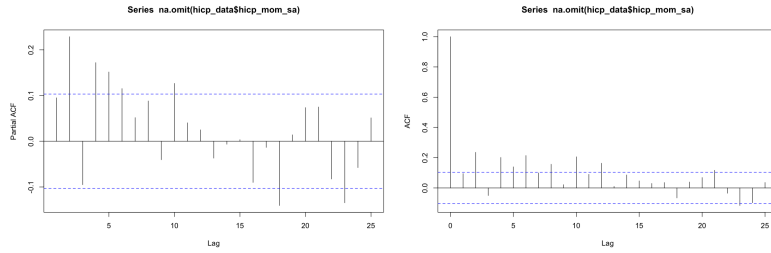


Figure 4: ACF and PACF of Month-on-Month (m-o-m) HICP Inflation

For the month-on-month (m-o-m) HICP inflation rate, the empirical ACF plotted in Figure 4 exhibits significant initial short-run lags but undergoes a relatively rapid decay, showing that the log-first-differenced transformation can be stable, stationary series. The corresponding PACF displays statistically significant spikes at lags 1, 2, and 4, without any of the long-memory or random walk characteristics compared to y-o-y data. Minimization of the Bayesian Information Criterion (BIC) isolates an $ARMA(4, 2)$ specification as the optimal model for m-o-m HICP inflation. For the national CPI robustness check, the BIC isolates a $ARMA(1, 1)$ model.

Diagnostic of the m-o-m HICP model shows specific estimation challenges. When the Ljung-Box test is conducted using a short horizon of 12 lags, the null hypothesis of no serial correlation cannot be rejected (p -value = 0.2407), establishing the short-run white noise properties of the residuals. However, when the diagnostic horizon is extended to 24 lags, the test rejects the white noise null hypothesis at the 5% level (p -value = 0.02926). This residual correlation shows residual seasonality that was not fully captured by the TRAMO-SEATS procedure. This empirical issue can be driven by the price volatility during the COVID-19 pandemic and the later 2022 energy crisis, which created significant spikes at lags 23 and 25 in the residuals. The diagnostic plots and residual properties for the m-o-m HICP series are provided in the appendix (see Figure 19).

2.3 Univariate Forecasts

Using the univariate models, projections were generated to evaluate the baseline trajectory of the German economy through 2026 and 2027.

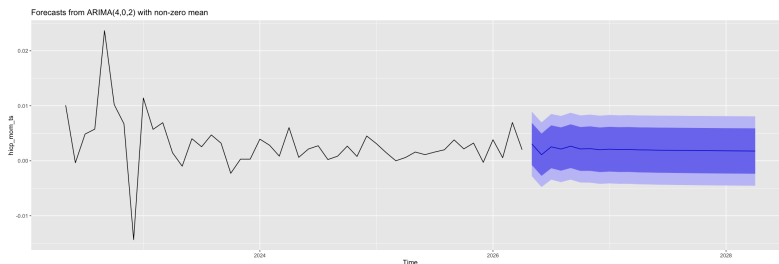


Figure 5: Univariate Forecast for Month-on-Month HICP Inflation

The projection for month-on-month HICP inflation is displayed in Figure 5. For the immediate horizon, the model projects a modest short-term price increase of 0.31%. However, due

to the high historical variance embedded in the monthly data, the standard confidence intervals are wide. Over the medium-term horizon, after absorbing short-run cyclical fluctuations in late 2026, the model projects that monthly inflation will stabilize smoothly between 0.18% and 0.20%. This stable monthly path translates into an implied annualized headline inflation rate of approximately 2.2% to 2.4%, slightly above the ECB’s explicit target.

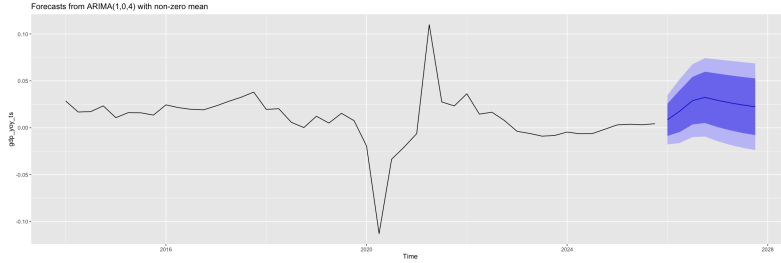


Figure 6: Univariate Forecast for Year-on-Year (y-o-y) Real GDP Growth

The forecast generated by the $ARMA(1,4)$ model for year-on-year Real GDP growth is presented in Figure 6. The forecast path suggests that the dynamics of year-on-year economic growth will gradually accelerate throughout 2026. The expansion is projected to reach a peak in the final quarter of 2026 (2026Q4), achieving a projected annual growth rate of 3.2%. Following this peak, the model projects a gradual deceleration over the course of 2027, with growth rates returning toward 2.2%.

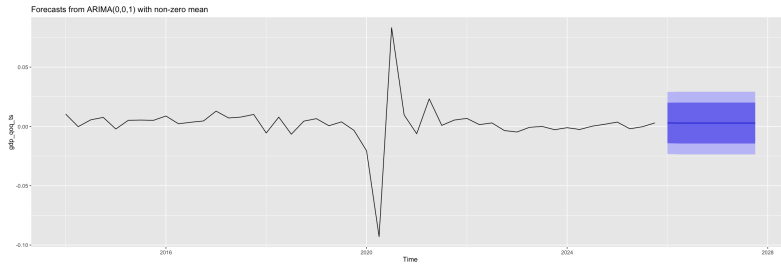


Figure 7: Univariate Forecast for Quarter-on-Quarter (q-o-q) Real GDP Growth

Furthermore, the forecast generated from the quarter-on-quarter (q-o-q) GDP growth shows a different trajectory, as seen in Figure 7. Because the q-o-q series closely resembles a white noise process and contains only a first-order moving average component, the historical memory of the model is highly short-lived. Consequently, from the second quarter of 2026 onward, all moving average dynamics completely disappear. The forecast path drops into a flat, horizontal line, projecting an unvarying growth rate equal to the historical sample mean of 0.28% per quarter. When annualized, this quarterly rate implies a steady growth pace of approximately 1.1%.

A critical comparative analysis of these two univariate GDP forecasts reveals a significant discrepancy between their implied long-run growth paths, with the y-o-y model converging to 2.2% and the q-o-q model settling at 1.1%. This structural divergence can be a consequence of how each transformation processes historical volatility and extreme shocks. Year-on-year data are inherently smoother because the rolling twelve-month window filters out high-frequency noise, allowing the model to better capture and preserve long-term expansionary trends. Conversely, quarter-on-quarter data are highly volatile and react intensely to sharp, short-run macroeconomic shocks. The extreme, negative shocks recorded during the COVID-19 lockdowns and the

2022 energy crisis led to pronounced downward spikes in the q-o-q series. Because these shocks are unweighted and independent in a quarter-on-quarter model.

3 Multivariate Models: Vector Autoregressive (VAR) Analysis

3.1 Exogenous Variables and Monetary Policy Proxy

While univariate formulations provide useful statistical baselines, they are fundamentally limited because they ignore the structural interactions and joint feedback mechanisms that drive the macroeconomic system. To overcome these limitations, we extend our analysis to a multivariate framework by incorporating two exogenous transmission channels that historically drive the German business cycle: an empirical proxy for the stance of monetary policy and an indicator of global commodity cost shocks.

The first expansion involves constructing a continuous and structurally consistent short-term interest rate series to capture monetary policy. This is empirically challenging because Germany transitioned from the independent monetary policy of the Deutsche Bundesbank to the centralized framework of the European Central Bank (ECB) on January 1, 1999, which coincided with the launch of the Euro. Furthermore, the overnight interbank market benchmark, the Euro Over Night Index Average (EONIA), was only formally established in 1999. To extend the interest rate series back to the beginning of our multivariate sample, data prior to 1999 are reconstructed using a synthetic series that matches the historical German money market rates back to January 1, 1994.

A second empirical obstacle emerged following the 2008 global financial crisis, as the ECB lowered its main refinancing rate to zero, reaching the Zero Lower Bound (ZLB). During this ZLB period, conventional short-term interest rates became flat and failed to reflect the expansionary stance of unconventional monetary policies, such as asset purchase programs and targeted long-term refinancing operations. To accurately capture the true stance of monetary policy when conventional rates are constrained, we utilize the shadow interest rate methodology developed by [Wu and Xia \(2016\)](#), extracted directly from Cynthia Wu’s research repository. For the ZLB period spanning from the third quarter of 2008 (2008Q3) through the second quarter of 2022 (2022Q2), the flat conventional rates are replaced with these unconstrained shadow interest rates. Following the return of high inflation in mid-2022, when the ECB aggressively hiked policy rates well above the zero floor, the series reverts to standard short-term money market rates, which replaced EONIA.

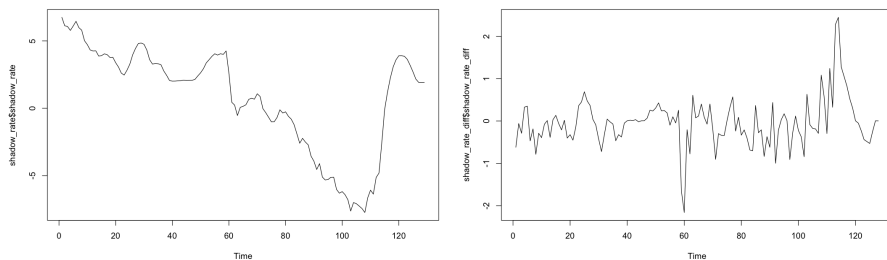


Figure 8: Historical Stance of the Proxy Interest Rate (Levels and First Differences)

The resulting synthesized monetary policy series is plotted in Figure 8, displaying the his-

torical path in levels (left panel, expressed as a percentage) and in first differences (right panel) from January 1, 1994, to January 1, 2026. The shadow rate successfully captures the deep negative territory reached during the peak QE years, dropping below -5% in the late 2010s, before exhibiting a sharp reversal during the post-pandemic inflationary spike. Formal ADF unit root tests indicate that the synthesized interest rate series is non-stationary when evaluated in levels. Despite this finding, we maintain the interest rate variable in levels within our baseline VAR models. Econometric theory and practice show that differencing an interest rate variable removes essential long-run level information regarding the monetary stance, while interest rates themselves are theoretically bounded and do not follow an unconstrained deterministic trend.

The second exogenous channel incorporated is a cost-push commodity shock, represented by the global spot price of Brent crude oil. Given Germany's position as a highly industrialized, manufacturing-intensive economy that lacks significant domestic mineral reserves, its industrial sector is highly sensitive to external energy price shocks. The spot price of Brent crude oil is sampled in USD per barrel, covering an extensive historical timeline from May 1, 1987, to March 1, 2026.

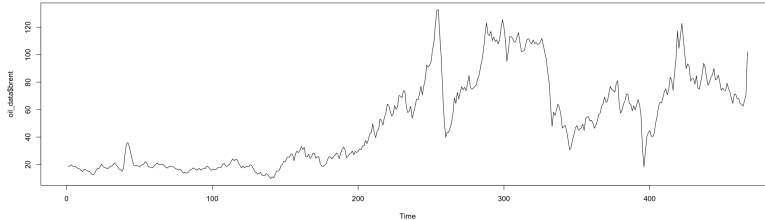


Figure 9: Historical Evolution of the Brent Crude Oil Spot Price

The historical trajectory of Brent crude oil is illustrated in Figure 9, highlighting its extreme historical volatility, marked by massive collapses during global demand downturns (2008 and 2020) and massive surges during supply disruptions (2022). To align these monthly indicators with our quarterly macroeconomic core, the monthly interest rate and log oil price series are converted to a quarterly frequency by calculating the simple arithmetic average across each three-month block. To ensure maximum internal consistency across the stationary VAR specification, the core macroeconomic indicators (Real GDP and HICP inflation) are modeled using their first-differenced log forms to represent quarter-on-quarter changes, which are subsequently annualized for better economic interpretation.

3.2 VAR Specification and Diagnostics

We construct a baseline reduced-form Vector Autoregressive (VAR) model where the current value of each endogenous variable is explained solely by its own historical lags and the historical lags of all other variables in the system.

The model contains four variables: log-differenced Brent oil price, annualized q-o-q Real GDP growth, annualized q-o-q HICP inflation, and interest rate in levels.

To determine the optimal lag length for the system, formal statistical selection criteria are computed. The Akaike Information Criterion (AIC) and related lag-selection tests recommend to use 2 lags. To test for residual serial correlation, a Portmanteau test is applied across the system. The test yields a p-value of 0.2555, thus failing to reject the null hypothesis of no

serial correlation and confirming that 2 lags are statistically sufficient to capture the system’s dynamic persistence. Furthermore, the structural stability of the estimated VAR is confirmed by computing the roots of the characteristic polynomial. All estimated roots fall strictly inside the unit circle, with the maximum modulus found to be 0.9259, ensuring that the system is stable.

3.3 Identification and Impulse Responses

To achieve identification, we utilize a recursive structural framework by applying a Cholesky decomposition to the reduced-form model. This method establishes a strict causal chain where variables ordered higher have an immediate impact on lower variables, but lower variables can only affect higher variables with a time lag. Our chosen structural ordering is arranged as follows:

$$\text{Oil Price, Real GDP Growth, HICP Inflation, Interest Rate} \quad (1)$$

The global spot price of oil is placed first as it represents an exogenous supply and cost-push shock driven by international commodity markets. Real GDP growth is ordered second, reflecting the hypothesis that aggregate output changes occur more rapidly than price level adjustments in response to shocks. HICP inflation is positioned third, capturing the presence of short-run nominal rigidities—such as menu costs or long-term contracts—which prevent consumer prices from responding instantaneously to monetary changes. Finally, the proxy interest rate is placed last, functioning as the fully endogenous reaction function of the central bank, which observes and responds contemporaneously to all other innovations in the system.

To evaluate how these structural shocks propagate through the German economy, Impulse Response Functions (IRFs) are computed over an extended horizon. We initially compare the IRF profiles across different lag lengths ($p = 2$, $p = 4$, and $p = 6$) to test for sensitivity to lag selection.

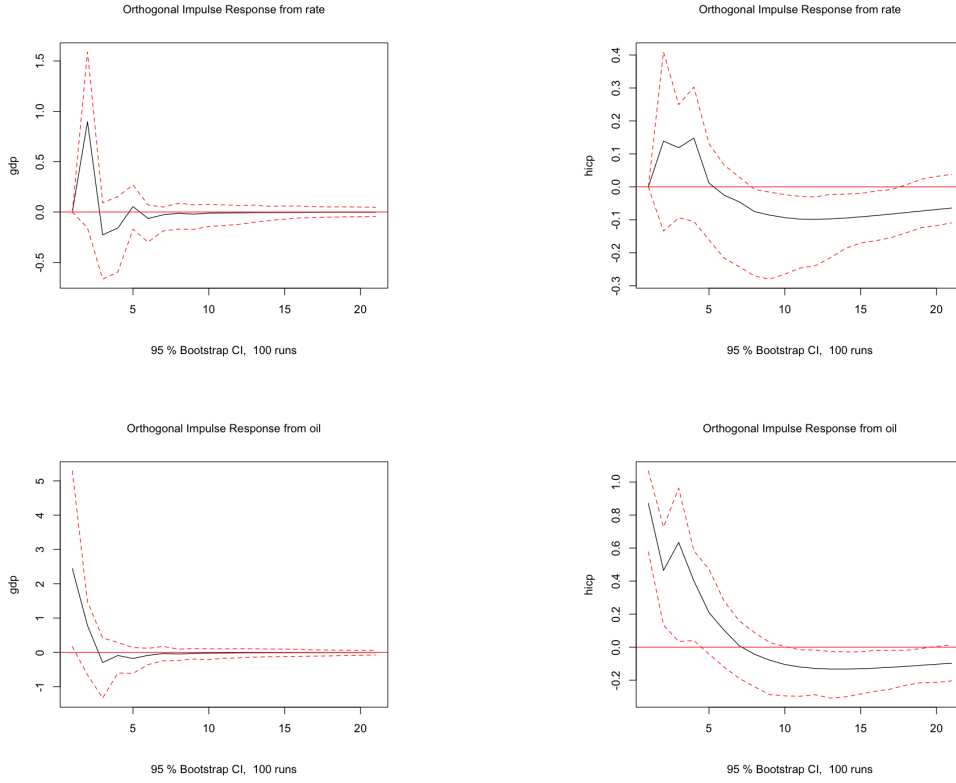


Figure 10: Impulse Response Functions for the Baseline $VAR(2)$ Specification

The impulse responses generated by the baseline $VAR(2)$ specification are presented in Figure 10. An empirical finding is that the 2-lag model exhibits a “price puzzle”: following a restrictive, positive monetary policy shock, the price level displays an immediate, counterintuitive increase rather than a contractionary decline. Furthermore, the monetary policy shock has almost no effect on real economic output. To capture longer transmission lags and more accurately reflect the annual business cycle, a model with a deeper lag structure is estimated. Extending the system to 6 lags to capture potential long-term transmission dynamics did not yield any material improvements or structural changes. Consequently, a $VAR(4)$ specification incorporating 4 lags is selected as our primary multivariate framework.

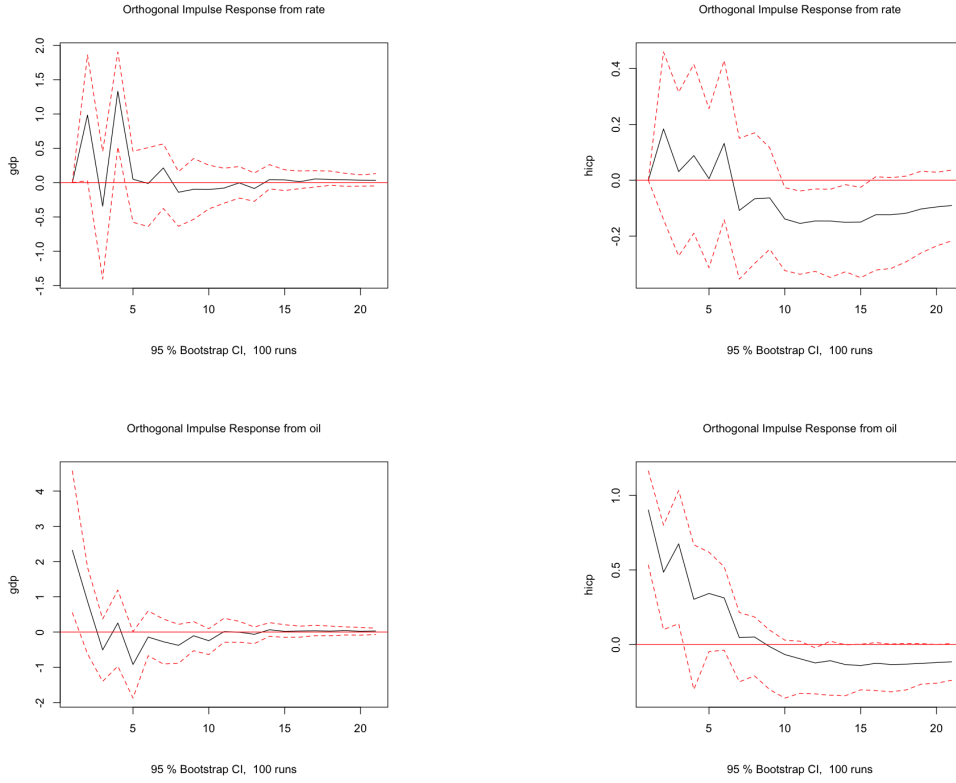


Figure 11: Impulse Response Functions for the Primary $VAR(4)$ Specification

The structural impulse responses for the primary $VAR(4)$ model are showed in Figure 11. While the short-run price puzzle remains partially visible in the immediate quarters following a monetary policy shock, the 4-lag model reveals that inflation has a statistically significant decline over the medium-term horizon. Crucially, even with 4 lags, there is a complete absence of a statistically significant drop in real GDP growth following a monetary tightening. In standard macroeconomic theory, monetary policy achieves disinflation by dampening aggregate demand, cooling economic activity, and opening a negative output gap. The complete absence of an output contraction in Germany can be interpreted through the anchored inflation expectations.

Examining the transmission of commodity shocks, a positive shock to the price of oil triggers a statistically significant, immediate surge in domestic HICP inflation, as external energy costs pass rapidly into consumer prices. Counterintuitively, the impulse responses show that a positive oil price shock also have a positive impact on German Real GDP growth. Standard economic theory intuition is that an increase in energy costs acts as a negative supply shock that compresses corporate profit margins and reduces aggregate output. This positive output paradox can be resolved by analyzing the underlying drivers of global commodity prices. If global oil prices are driven upward by strong international demand as the massive economic expansions in emerging Asian economies recorded during the 2003–2006 period—global aggregate demand rises. As demonstrated by [Khalil \(2020\)](#), highly export-oriented economies with large manufacturing sectors benefit significantly from these demand-driven commodity shocks. The surge in international demand for high-value industrial exports outweighs the domestic cost-push burden of higher energy imports. This specific structural mechanism explains why a pro-export industrial country like Germany can experience a temporary expansion in real GDP growth alongside

rising oil prices.

As an alternative structural check, an SVAR model with parameter restrictions was estimated. This alternative specification assumes that the central bank cannot respond instantaneously to global oil price shocks within the same quarter. The economic rationale is that monetary authorities primarily monitor core inflation measures. However, a formal statistical test of these overidentifying structural restrictions rejects the null hypothesis, indicating that this restricted framework is incompatible with the historical data. The empirical impulse responses generated by this restricted SVAR model differ minimally from our baseline recursive SVAR model.

3.4 VAR Forecasts

Using the primary 4-lag recursive SVAR model, out-of-sample forecasts were generated to project the path of the German economy through 2026 and 2027.

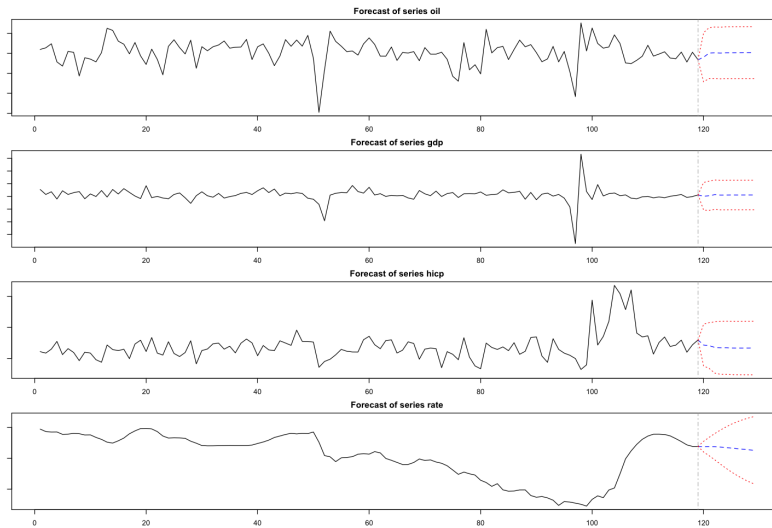


Figure 12: Out-of-Sample Forecasts from the Primary $VAR(4)$ Model

The joint projections for all four system variables are presented in Figure 12. For the interest rate, the model predicts an initial value of 1.84% in the first quarter of 2026 (2026Q1). The path is projected to increase gradually, reaching a peak of 2.03% in the third quarter of 2026 (2026Q3), before returning on a sustained downward trajectory that drops to a long-run value of 1.14% by the final quarter of 2028 (2028Q4).

For headline HICP inflation, the VAR forecast projects a classic macroeconomics “soft landing”. In the immediate forecast horizon, annualized inflation remains slightly elevated at 2.36%. However, it declines in subsequent quarters within a corridor between 1.42% and 1.59% from the fifth quarter onward. It is critical to note that this projection does not incorporate unpredicted geopolitical escalations, such as potential supply-side conflicts in the Middle East.

For economic activity, the model projects an immediate contraction in Real GDP growth of -1.29% in the first quarter of the forecast horizon. This short-run drop represents the lagged transmission of past restrictive monetary policy actions and energy cost shocks that cooled the economy in late 2025. Following this initial dip, the economy is projected to rebound, returning to a stable and modest expansionary path ranging between 1.0% and 1.7% growth. Finally, the projection for the price of oil exhibits wide confidence intervals, spanning from $-\$154$ to $+\$97$

per barrel. The point estimate projects a sharp initial decline of -28.22% before stabilizing around a growth rate of $+3\%$ to $+5\%$ per year.

4 Bayesian VAR and Sign Restrictions

4.1 Bayesian Estimation

The Bayesian framework treats the parameters as random variables and shrinks them toward a prior distribution. We employ a standard Minnesota prior distribution, which assumes that the best description of an economic variable's future value is its current value, shrinking the system toward a random walk framework.

To implement BVAR, we utilize a hierarchical prior structure that treats the hyperparameters as random variables to be estimated from the data. The global tightness hyperparameter, λ , governs the weight placed on the prior distribution relative to the raw sample information. A value of $\lambda \rightarrow 0$ implies that the model relies entirely on the prior restrictions, while $\lambda \rightarrow \infty$ collapses the BVAR into an unconstrained OLS estimation. Through hierarchical optimization, the optimal value of λ is estimated to be 0.29, where the trace plot for λ exhibits fluctuation around the 0.20 to 0.40 band. The lag decay hyperparameter, α , which controls the speed at which more distant historical lags are shrunk toward zero, is fixed at a value of $\alpha = 2.0$.

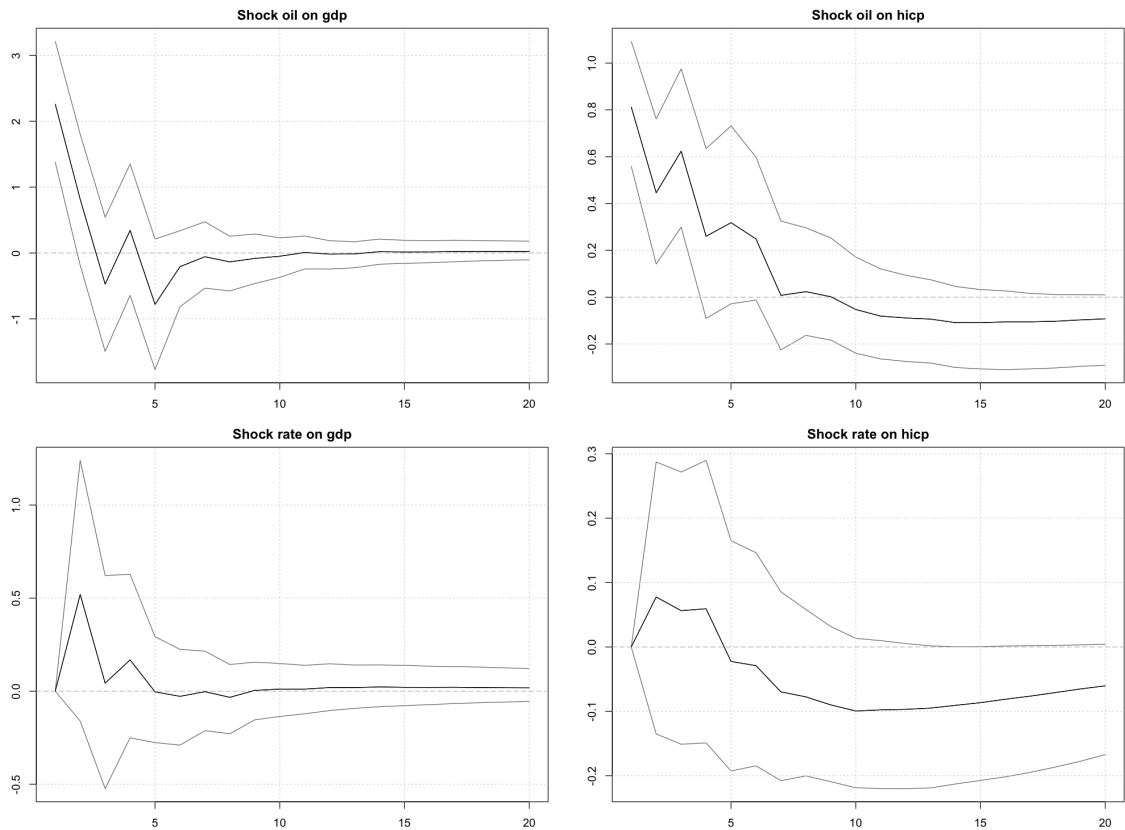


Figure 13: Impulse Responses from BVAR Model

The structural impulse responses are showed in Figure 13. The estimated BVAR trajectories match the VAR model, but it is producing smoother transitions between the fifth and tenth quarters. Furthermore, the 95% posterior credible intervals for the impact of an oil shock on

HICP inflation are tighter and more precise compared to baseline VAR. For a monetary policy tightening, the interest rate shock exhibits smooth decay. However, it fails to resolve the short-run price puzzle. the response of HICP inflation to an unexpected interest rate hike remains strictly positive in the short run, and its 95% credible interval does not cross below zero during the initial quarters.

4.2 BVAR Sign Restrictions and Structural Identification

To directly resolve the empirical price puzzle, we impose explicit theoretical macroeconomics sign restrictions onto the Bayesian VAR framework. This identification method restricts the signs of the impulse responses for specific variables over a short horizon, allowing us to isolate structural shocks based on economic theory without imposing restrictions on the contemporaneous coefficients. The structural identification scheme restricts three shock, leaving a fourth shock fully unidentified. (see Table 2)

Table 2: Identification of Structural Shocks via Sign Restrictions

Variable	Demand Shock	Oil Shock	Monetary Policy Shock
Oil Price (oil)	0 (NA)	+1	0 (NA)
Economic Growth (gdp)	+1	-1	-1
Inflation (hiep)	+1	+1	-1
Interest Rate (rate)	+1	+1	+1

The economic rationale for these sign restrictions is derived from a standard New Keynesian macroeconomics framework. A positive aggregate demand shock is defined as an innovation that drives an expansion in economic growth (+1), consumer price inflation (+1), and short-term interest rates (+1), while its immediate impact on the global price of oil is left unconstrained. A positive commodity supply shock (a adverse cost-push oil shock) is defined as an shock that increases the price of oil (+1), increases consumer price inflation (+1), and triggers a monetary tightening (+1), while forcing a contraction in real economic growth (−1). This supply restriction contrasts with the empirical results of our unconstrained models, forcing the model to evaluate a negative output effect. Finally, a contractionary monetary policy shock is identified by restricting the short-term interest rate to rise (+1), while forcing an immediate structural contraction in both real GDP growth (−1) and HICP inflation (−1). By explicitly restricting inflation to decline, the price puzzle is eliminated by construction.

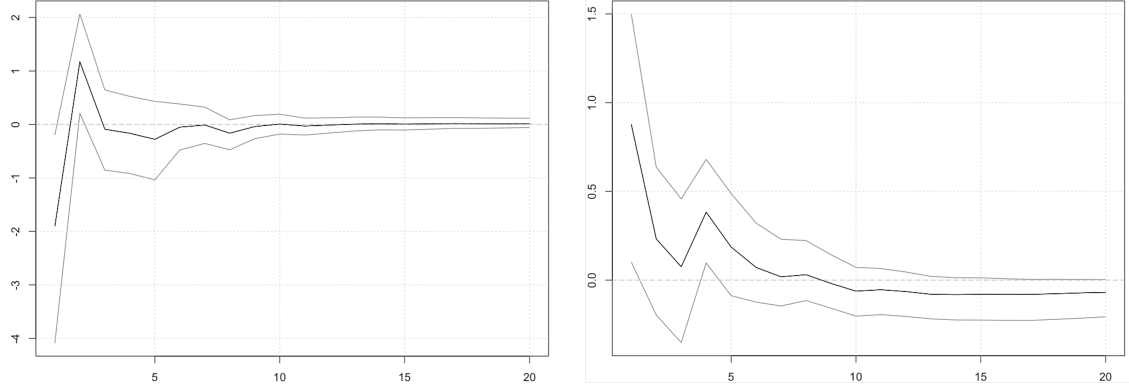


Figure 14: Impulse Responses under Sign Restrictions: Oil Shock (Left: Response of GDP, Right: Response of HICP)

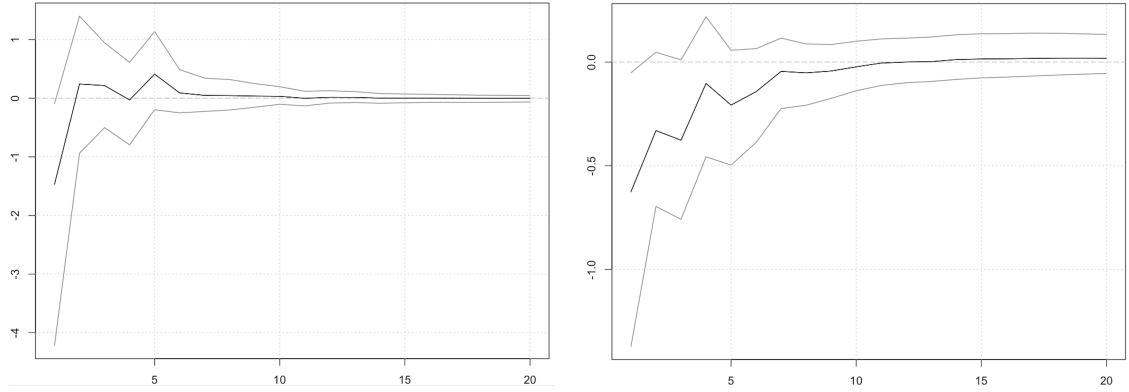


Figure 15: Impulse Responses under Sign Restrictions: Interest Rate Shock (Left: Response of GDP, Right: Response of HICP)

The structural impulse responses generated under the sign restrictions identification scheme are presented in Figure 14. Evaluating the oil supply shock, we observe that the restriction forces a negative output response. While the unconstrained VAR yielded a counterintuitively positive output effect due to global demand factors, the sign-restricted model isolates a negative growth trajectory, though the response is not statistically significant.

Looking at the monetary policy shock in Figure 15, the explicit restriction ensures that the price puzzle disappears, forcing inflation to decline immediately following an interest rate hike. However, because the historical data do not strongly support an immediate price contraction, the sign-restricted negative inflation response are statistically insignificant over the entire horizon.

4.3 BVAR Forecasts

Forecasts were generated using BVAR model to evaluate the impact on our macroeconomic predictions.

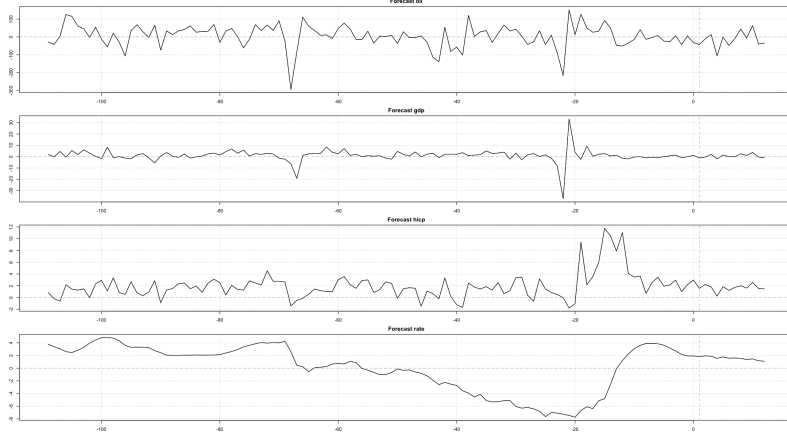


Figure 16: Out-of-Sample Macroeconomic Forecasts from the BVAR Model

The projections from the BVAR model are displayed in Figure 16, plotting the point estimates without their wide intervals to maintain visual clarity. Because the Bayesian framework incorporates broad parameter uncertainty, intervals are wider than baseline VAR confidence bands, meaning these forecasts must be interpreted with caution. The proxy short-term interest rate exhibits a very gradual, smooth downward trajectory, mirroring the baseline VAR profile. For Real GDP growth, the BVAR model projects a short-term contraction over the upcoming year, followed by a robust medium-term recovery that drives growth up toward 3% after two and a half years. For consumer prices, the BVAR projects a sustained disinflationary path that pushes inflation below the ECB’s 2% target, with a gradual return to equilibrium occurring in the third year of the forecast horizon.

5 Models with Trend Variables

5.1 VAR in Levels Specification and Analysis

To evaluate the empirical validity of our differencing transformations, we estimate alternative macroeconometric models that retain the non-stationary trend components of the data. The first robustness check involves estimating an unconstrained VAR model in levels. By modeling the variables in levels, we allow the VAR system to implicitly capture potential cointegrating relationships without imposing explicit cointegration vectors. For this specification, Real GDP and the HICP price index are included in their seasonally adjusted log-level forms, the spot price of Brent oil is log-transformed, and the interest rate is maintained in its raw level form. To maintain comparability with our baseline model, we preserve the same ordering: log oil price, log GDP, log HICP, and the interest rate.

The optimal lag structure for the levels model is evaluated using the `VARselect` procedure, which recommends either 2 or 3 lags. We initially estimate a $VAR(2)$ specified in levels and evaluate its residuals. A Portmanteau test applied to the 2-lag model rejects the null hypothesis of no serial correlation (p -value = 0.0254), indicating that 2 lags are insufficient to capture the dynamic persistence of the non-stationary series. Extending the system to 3 and 4 lags successfully eliminates residual autocorrelation. To align with quarterly data properties (covering a full one-year historical window) and to ensure a direct comparison with our baseline stationary model, we adopt a $VAR(4)$ specified in levels. This annual lag structure better captures long-run

cointegrating dynamics.

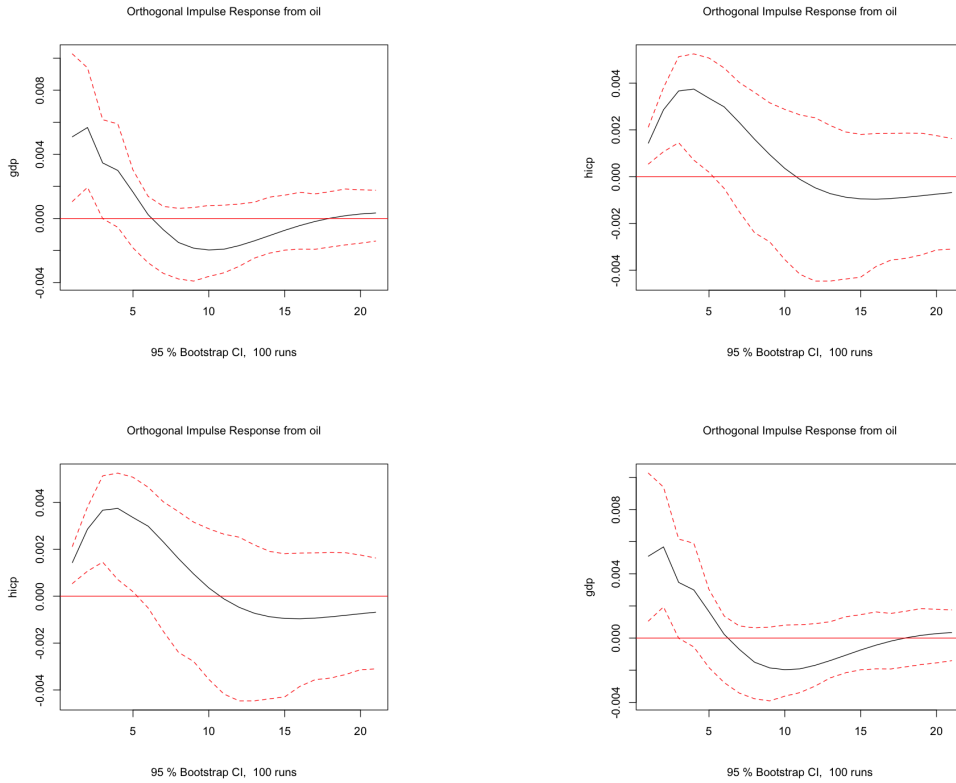


Figure 17: Impulse Responses from the $VAR(4)$ Specified in Levels

The structural impulse responses for the $VAR(4)$ in levels are displayed in Figure 17. The transmission of an oil price shock, the levels model matches our baseline results, generating a positive output expansion in Real GDP alongside a sustained increase in the HICP price level. The price level response reaches its peak in the fourth quarter before gradually goes down, while the output expansion decays more rapidly. Looking at monetary policy, the levels model reveals a significant structural shift: the price puzzle disappears, as an unexpected interest rate hike no longer triggers a counterintuitive spike in inflation. However, the interest rate shock does not exert a statistically significant contractionary effect on either the log-level of GDP or the log-level of HICP. This lack of significance reflects a substantial loss of estimation efficiency. Because the variables are non-stationary and do not incorporate explicit cointegration constraints, the estimated parameter uncertainty is high, producing wider confidence intervals that reduce the statistical power of the IRFs.

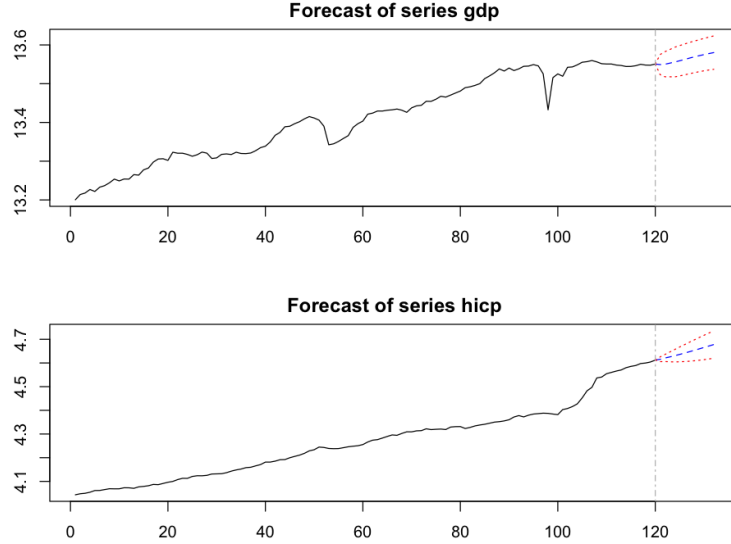


Figure 18: Long-Run Out-of-Sample Forecasts from the VAR in Levels Model

The out-of-sample forecasts generated by the VAR in levels are illustrated in Figure 18. The forecast paths track the long-run historical trends of the German economy. The model projects a very stable and realistic long-run equilibrium, with consumer prices tracking a smooth path that equates to an average inflation rate of 2.06% per year. Simultaneously, Real GDP is projected to expand at a steady pace of approximately 1% per year in real terms.

5.2 Vector Error Correction Model (VECM)

The second trend-based robustness check involves estimating a Vector Error Correction Model (VECM) to explicitly model the long-run cointegrating vectors and short-run adjustment dynamics. To determine the exact number of stable cointegrating relationships among our non-stationary variables, a formal Johansen cointegration procedure is executed. When the trace and maximum-eigenvalue statistics are evaluated at a conventional 5% significance level, the test fails to find evidence of a stable cointegrating vector. For the null hypothesis of zero cointegrating vectors ($r = 0$), the empirical test statistic is 51.02, which fails to exceed the critical value threshold of 62.99.

To maintain maximum consistency with our previous level specifications, we estimate a VECM using the same level transformations, 4 autoregressive lags, and incorporate a deterministic trend within the cointegration space (`ecdet = "trend"`). Given the statistical absence of a natural cointegrating vector, we manually impose a single cointegrating relationship ($r = 1$) to calculate the impulse responses and evaluate the VECM framework. The structural impulse responses for the VECM are reported in the appendix (see Figures 20 to 23). Because there is no statistical evidence supporting a stable long-run cointegrating relationship, the VECM specification is structurally weak. Out-of-sample forecasts generated from the VECM closely mirror the trajectories of the levels model, and the VECM forecast graphic is provided in the appendix (see Figure 24).

6 Conclusion

This term paper conducted a comprehensive macroeconometric evaluation of forecasting and structural shock propagation in Germany, utilizing an array of univariate and multivariate models. In the univariate framework, optimized ARIMA specifications provided stable baseline short-term projections, though they revealed a significant divergence between quarter-on-quarter and year-on-year transformations regarding long-run implied steady states. This divergence highlights the sensitivity of unvaried models to historical macroeconomic volatility and asymmetric shocks.

In the multivariate framework, a recursive Structural VAR (SVAR) identified via Cholesky ordering emerged as the most adequate and robust specification for short-to-medium term forecasting. The model predicts inflation below 2% and Real GDP growth recovering toward a modest trend following a brief initial dip. These baseline conclusions remained robust when evaluated against an alternative VAR specified in levels

Our structural analysis revealed an empirical features of German economy. First, unconstrained monetary policy shocks fail to trigger a significant contraction in real economic output. Second, positive global oil price shocks generate a temporary expansion in German Real GDP growth. This output paradox is explained by the demand-driven nature of international commodity markets, where rising global demand drives up energy prices while simultaneously boosting foreign demand for German industrial manufacturing exports. Finally, while advanced Bayesian VAR formulations provided smoother impulse responses, the forecast had wide probability intervals.

Appendix

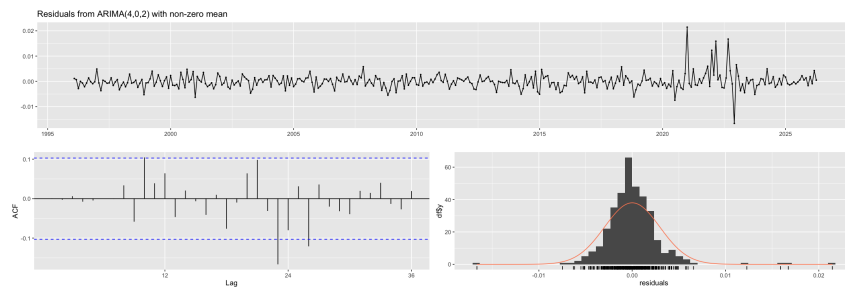


Figure 19: Detailed Residual Diagnostics for the Month-on-Month HICP $ARMA(4, 2)$ Model

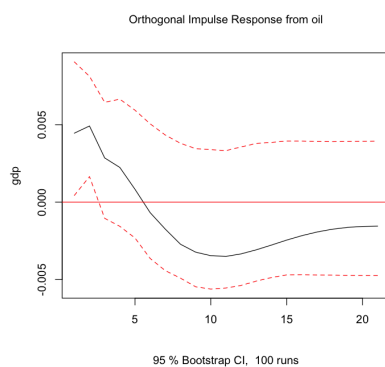


Figure 20: VECM Impulse Response: Oil Price Shock to Real GDP

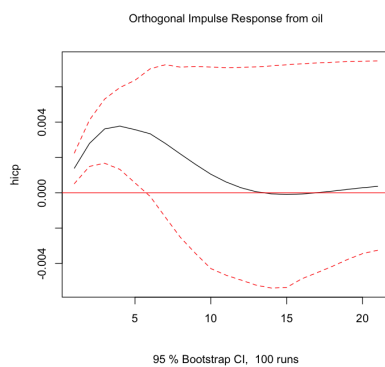


Figure 21: VECM Impulse Response: Oil Price Shock to HICP Inflation

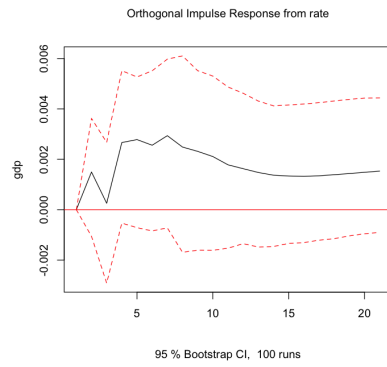


Figure 22: VECM Impulse Response: Interest Rate Shock to Real GDP

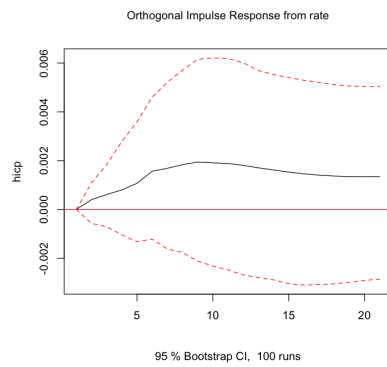


Figure 23: VECM Impulse Response: Interest Rate Shock to HICP Inflation

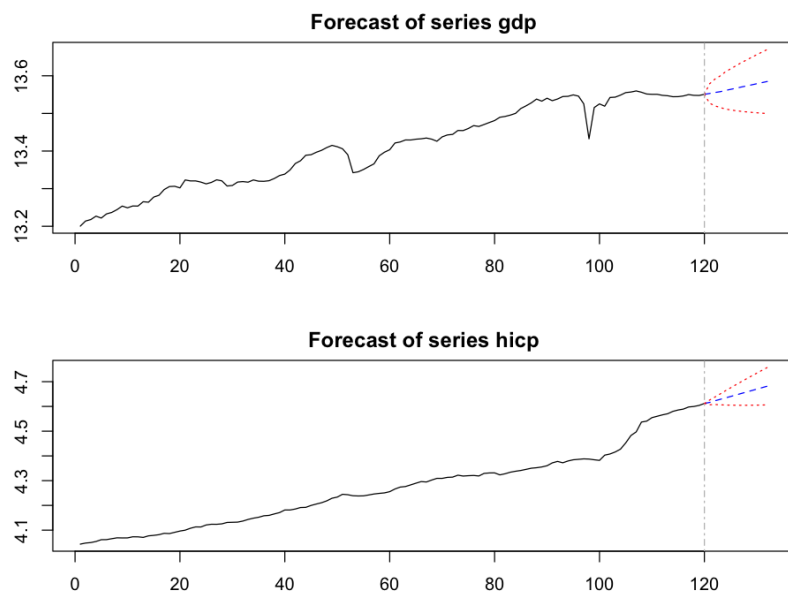


Figure 24: Out-of-Sample Macroeconomic Forecasts from the VECM Specification

References

- Deutsche Bundesbank (2019). *Dynamics of inflation in Germany and the Euro Area: Structural drivers and monetary implications*. Monthly Report, November 2019.
- Deutsche Bundesbank (2022). *The impact of the energy crisis on the German industrial sector and potential output*. Monthly Report, October 2022.
- Deutsche Bundesbank (2023). *The transmission of commodity price shocks into German consumer prices*. Discussion Paper No. 14/2023.
- Dustmann, C., Bernd, F., Schönberg, U., & Spitz-Oener, A. (2014). From Sick Man of Europe to Economic Superstar: Germany's Resurgent Economy. *Journal of Economic Perspectives*, 28(1), 167–188.
- European Central Bank (2010). *The monetary policy transmission mechanism during the financial crisis*. ECB Monthly Bulletin, May 2010.
- Jarausch, K. H. (1994). *The Rush to German Unity*. Oxford University Press.
- Khalil, M. (2020). Commodity Price Shocks and the Manufactured Goods Exporter: The Case of Germany. *Bundesbank Discussion Paper*, No. 42/2020.
- Krippner, L. (2015). *Zero Lower Bound Term Structure Models: A Practitioner's Guide*. Palgrave Macmillan.
- Sinn, H.-W. (2002). Germany's Economic Problems. *CESifo Working Paper*, No. 754.
- Wu, J. C., & Xia, F. D. (2016). Measuring the Macroeconomic Impact of Monetary Policy at the Zero Lower Bound. *Journal of Money, Credit and Banking*, 48(2-3), 253–291.